INCH-POUND
MIL-PRF-45307C(AT)
13 November 1996
SUPERSEDING
MIL-P-45307B(AT)
7 April 1989

PERFORMANCE SPECIFICATION

PUMP, HYDRAULIC, AXIAL PISTONS: MANUAL, WITH FIRE-CONTROL SWITCH

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hydraulic pump assembly consisting of an axial piston pump, a control handle, and a fire-control switch. The nominal displacement of the pump is 0.80 cubic inch ($in.^3$) [13 cubic centimeters (cm^3)] per revolution (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2520

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-H-46170 - Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

ARMY

8674351	- Housing Assembly.
10911437	- Handle Assembly.
11654400	- Pump Assembly, Manual Elevation.
11654401	- Pump.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4	- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).
ANSI/SAE AS478	- Identification Marking Methods (DoD Adopted).

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Zinc Coating, Electrodeposited (DoD Adopted).
ASTM D522	- Standard Test Method for Mandrel Bend Test of Attached Organic Coatings.

(Application for copies should be addressed to American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

GENERAL MOTORS CORPORATION

GM 9540P	- Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE AS 1933	- Age Control of Age Sensitive Elastomeric Material (DoD Adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Unless otherwise specified herein or specified in the applicable drawings, materials used shall be in accordance with the manufacturer's materials specifications for semitrailers. The materials shall be capable of meeting all the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable. Materials and special treatments shall be utilized to prevent corrosion and to resist harmful effects of dust, water and fungi. Dissimilar metals shall not be used in intimate contact without protection from galvanic corrosion (see 3.2.5 and 4.6.1).

3.2.1 Hazardous materials. Asbestos, Cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item (see 4.6.1).

3.2.2 Corrosion protection. The vehicle shall be fabricated from compatible materials providing corrosion protection and coating adherence equal to or exceeding that provided by hot dip galvanized 1010 steel with a minimum coating thickness of 2 mils (50 micrometers) on steel sheet less than 0.0625 inches (in.) thick, 2.5 mils on steel sheet greater than 0.0625 in. thick or 0.75 mil (19 micrometers) on pre-galvanized steel sheet 0.0625 in. thick or less. A material sample shall be capable of meeting or exceeding the corrosion resistance provided by a galvanized sample (as described above) when subjected to the Mandrel Bend Test of ASTM D522, and followed by the Accelerated Corrosion Test of GM 9540P Method B, 120 cycles (see 4.6.1).

3.2.3 Metals. All metals used in the construction of the pump, except those which are in constant contact with the hydraulic fluid, shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion. The use of dissimilar metals, in intimate metal-to-metal contact, shall be avoided wherever practicable (see 4.6.1).

3.2.4 Age sensitive materials. The use of age sensitive materials shall be controlled in accordance with SAE AS 1933 (see 4.6.1).

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3.2.5 Anodizing. All aluminum-alloy parts shall be anodized in accordance with the contractor's specifications and shall meet all of the operational and environmental requirements specified herein (see 3.2 and 4.6.1).

3.2.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 3.2, 4.6.1 and 6.3.1).

3.3 Design and construction. The pump assembly shall be designed and constructed to meet the requirements of section 3. The pump assembly Drawing 11654400 mounting provisions shall be in accordance with Drawing 11654401. The external configuration of the pump assembly shall approximate the shapes of its three (3) major components; housing assembly Drawing 8674351, handle assembly Drawing 10911437 and pump Drawing 11654401 (see 4.4.1 and 4.6.1).

3.3.1 Lubrication. The pump shall be self-lubricating with no provision for lubrication other than circulating hydraulic oil from the normal hydraulic oil supply (see 4.4.1 and 4.6.1).

3.3.2 Displacement. The pump shall have a nominal displacement of 0.80 $in.^3$ (13 cm^3) per revolution (see 4.6.1).

3.3.3 Direction of rotation. The pump shall operate in either direction of rotation with maximum efficiency (see 4.6.2.1). Direction of flow shall reverse when shaft rotation is reversed (see 4.6.1).

3.3.4 Standard parts. Military standard parts shall be incorporated wherever possible. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification (see 4.4.1 and 4.6.1).

3.4 Performance. The pump shall be capable of satisfactory operation as specified herein, when using hydraulic oil as specified in 4.6.2.

3.4.1 Volumetric efficiency. The distance traveled by the piston rod of the cylinder shall not be less than 0.210 inch (in.) (0.53 centimeters (cm)) per revolution of pump (see 3.4.6, 3.4.7, and 4.6.2.1).

3.4.2 Hydraulic backlash. The hydraulic backlash shall not exceed 8 degrees (see 4.6.2.2).

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3.4.3 Proof pressure. The pump, including all parts and housing, shall withstand an internal pressure of not less than 3000 pounds per square inch (psi) [20 684 kiloPascals (kPa)] for 1 minute without damage to component parts (see 4.6.2.3).

3.4.4 Pump handle torque. The torque required to turn the pump handle shall not exceed 36 pound force inch (lbf. in) [4 Newton-meters (N-m)] (see 4.6.2.4.).

3.4.5 Pump leakage.

3.4.5.1 Low pressure. The pump shall show no evidence of external leakage when a pressure of 75 psi (517 kPa) is applied to a non-rotating pump and no more than 0.003 cubic inch per hour ($in.^3/h$) (0.05 cubic centimeter per hour (cm^3/h)) external leakage when the pump is operated (see 4.6.2.5).

3.4.5.2 High pressure. The pump shall show no evidence of external leakage when a pressure of 1500 psi (10 342 kPa) is applied to a non-rotating pump and no more than 0.003 $in.^3/h$ (0.05 cm^3/h) external leakage when the pump is rotated (see 4.6.2.5).

3.4.6 Low temperature. The pump assembly shall operate without special equipment at ambient temperatures down to -25 degrees Fahrenheit (°F) (-32 degrees Celsius (°C)) and shall meet the requirement specified in 3.4.1 after being subjected to a storage temperature of -65°F (-54°C) (see 4.6.2.6).

3.4.7 High temperature. The pump assembly shall operate without special equipment at ambient temperatures up to 125°F (52°C) and shall meet the requirement specified in 3.4.1 after being subjected to a storage temperature of 160°F (71°C) (see 4.6.2.7).

3.4.8 Shock. The pump shall show no evidence of damage or deformation and shall not demonstrate performance degradation after exposure to sawtooth shock pulses of 40 gravity units (g) for 11 milliseconds (ms) duration applied along each of the three major axes (see 4.6.2.8.)

3.5 Finish. The finish treatments of the external surfaces of the pump assembly is optional but shall meet all the performance and environmental requirements of this specification. However, the final finish paint color shall be white and conform to the applicable color chip number of FED-STD-595 (see table II and 6.2).

3.6 Marking. A nameplate conforming to ANSI/SAE AS478 shall be securely attached as indicated on Drawing 11654400. Identification markings shall include the following information (see 4.4.1):

a. Pump assembly
b. Manual elevation

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c. Manufacturer's identification
d. Serial number
e. Military part number
f. Federal stock number
g. Manufacturer's part number
h. Date of manufacture
i. Contract number
j. US

3.7 Workmanship. Workmanship shall be of a quality which shall assure a product free of burrs, rust, scratches, chips and sharp edges (see 4.4.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspections (see 4.4).

4.2 Inspections conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 18°F (23 ±10°C)
b. Barometric pressure: 28.5 (+2, -3) in. of mercury
[96.5 (+7, -10) kilopascals (kPa)]
c. Relative humidity: 50 ±30 percent (%)

4.2.1 Environmental tests. The pump assembly shall be operated at ambient conditions prior to proceeding with any of the environmental tests to obtain a baseline for determining satisfactory operation of the item during and after the test, as applicable.

4.3 First article inspection. When first article is required (see 3.1), inspection of first article sample(s) in accordance with ANSI/ASQC Z1.4 shall be conducted in accordance with table I (see 6.2).

4.4 Conformance inspection.

4.4.1 Examination. For examination purposes, samples (see 6.2) shall be inspected for defects specified in table II.

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance	
				Examination	Acceptance
Materials and construction	3.2 thru 3.3.4	4.6.1	X		
Defects	3.3, 3.3.1, 3.3.4, 3.5, 3.6 and 3.7	4.4.1	X	X	
Volumetric efficiency	3.4.1	4.6.2.1	X		X
Hydraulic backlash	3.4.2	4.6.2.2	X		X
Proof pressure	3.4.3	4.6.2.3	X		
Pump handle torque	3.4.4	4.6.2.4	X		X
Low pressure	3.4.5.1	4.6.2.5	X		X
High pressure	3.4.5.2	4.6.2.5	X		X
Low temperature	3.4.6	4.6.2.6	X		
High temperature	3.4.7	4.6.2.7	X		
Shock	3.4.8	4.6.2.8	X		

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Nonconformance in design and construction (see 3.3 and 3.3.1).	Visual
102	Dimensions affecting interchangeability out of tolerance (see 3.3 and 3.3.4).	SIE 1/
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor:		
201	Dimensions not affecting interchangeability out of tolerance (see 3.3).	SIE
202	Improper finish (see 3.5).	Visual
203	Improper marking (see 3.6).	Visual
204	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5 Acceptance tests. Samples selected for acceptance tests in accordance with ANSI/ASQC Z1.4 shall initially be examined in accordance with 4.4.1 and shall then be subjected to the tests specified in table I.

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4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 through 3.3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Performance. All performance testing shall be accomplished using hydraulic oil conforming to MIL-H-46170 (see 3.4).

4.6.2.1 Volumetric efficiency. To determine conformance to 3.3.3 and 3.4.1, the pump shall be connected to a double-acting cylinder with a piston area at each end of 2.75 square inches ($in.^2$) (17.7 square centimeters (cm^2)). A needle valve and a 2000 psi (13 788 kPa) gage shall be installed to each line from the pump to the cylinder. A pressure gage shall be installed between the needle valve and the pump port. The system shall be charged to a pressure of 1500 psi (10 342 kPa) with needle valve on the discharge side of the cylinder set to give a differential of 100 psi (689 kPa) between the ports of the pump. The control handle shall be rotated at a speed of one revolution per minute (rpm). Distance traveled by piston rod of cylinder shall be measured in one revolution of control handle by a dial indicator. This test shall be made once in a clockwise direction and once in a counterclockwise direction of rotation.

4.6.2.2 Hydraulic backlash. To determine conformance to 3.4.2, the degree of hydraulic backlash of the pump shall be determined by measurement with an indicator gage on the piston of the cylinder and a protractor indicator mounted on the case of pump with a pointer on the handle.

4.6.2.3 Proof pressure. To determine conformance to 3.4.3, an internal hydrostatic pressure of not less than 3000 psi (20 684 kPa) shall be applied to the pump and the pump port and housing for a period of 1 minute. The pump shall be examined for damage to component parts.

4.6.2.4 Pump handle torque. To determine conformance to 3.4.4, the pump shall be filled with hydraulic oil under a working pressure of 1500 psi (10 342 kPa) and shall move the piston with 60 psi (414 kPa) pressure differential, at a rate of rotation of not more than one rpm. The torque required to turn the handle shall be measured by an approved method.

4.6.2.5 Pump leakage. To determine conformance to 3.4.5.1 and 3.4.5.2, the pressures specified in 3.4.5.1 and 3.4.5.2 shall be applied to case port and both pressure ports while pump is not rotated and while pump is rotated at 10 to 20 rpm. The rate of external leakage shall be measured.

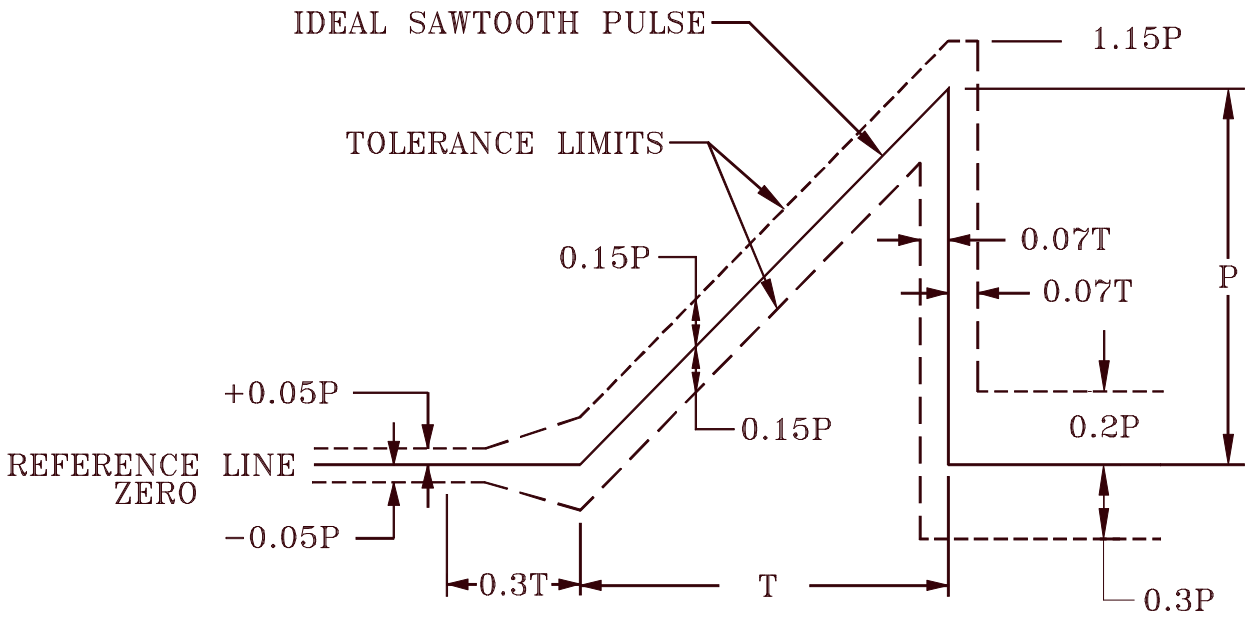
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4.6.2.6 Low temperature. To determine conformance to 3.4.6, the pump assembly shall be installed in a chamber in a manner that will simulate service usage. The internal chamber temperature shall be lowered to -65°F (-54°C), the storage temperature, and shall be maintained for a period of 24 hours. The internal chamber temperature shall be adjusted to -25°F (-32°C), the lowest temperature for which the pump assembly is designed to operate, and shall be maintained until temperature stabilization (see 6.3.3) of the pump assembly is reached. When raising or lowering the temperature of the chamber, the rate of temperature change during the test shall not exceed 18°F (10°C) per minute. The pump assembly shall be subjected to the test specified in 4.6.2.1. After testing, the pump shall be returned to 73 ±18°F (23 ±10°C) and subjected to the test specified in 4.6.2.1 (see 4.2.1).

4.6.2.7 High temperature. To determine conformance to 3.4.7, the pump assembly shall be installed in a chamber in a manner that will simulate service usage. The internal chamber temperature shall be raised to 160°F (71°C) and shall be maintained for a period of 48 hours while ensuring that the relative humidity is not in excess of 15%. The internal chamber temperature shall be adjusted to 125°F (52°C), the highest operating temperature for which the pump assembly is designed to operate, and shall be maintained until temperature stabilization of the pump is reached (see 6.3.3). When raising or lowering the temperature of the chamber, the rate of temperature change during the test shall not exceed 18°F (10°C) per minute. The pump shall be subjected to the test specified in 4.6.2.1. After testing, pump shall be returned to 73 ±18°F (23 ±10°C) and shall be subjected to the test specified in 4.6.2.1 (see 4.2.1).

4.6.2.8 Shock. To determine conformance to 3.4.8, the pump assembly shall be mounted by normal mounting provisions to a shock machine capable of producing the shock pulse shown in figure 1. Three sawtooth shocks of 40 g and time duration of 11 ms (see figure 1) in each direction shall be applied along three mutually perpendicular axes of the pump (total of 18 shocks). After the shock test, the pump assembly shall be examined for evidence of damage or deformation and performance degradation (see 4.2.1).

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NOTES:

1. Peak value (P) = 40 g.
2. Nominal duration (T) = 11 ms.
3. The oscillogram shall include a time about 3T long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and it's duration is T. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi + 0.1 Vi, where Vi is the velocity change associated with the ideal pulse which equals 0.5 PT. The integration to determine velocity change shall extend from 0.4T before the pulse to 0.1T after the pulse.

FIGURE 1. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump assembly covered by this specification is intended for use in combat vehicles for controlling movement of primary gun by manual operation, through its entire range of elevation and depression (see 1.1).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article inspection is required (see 3.1).
d. If finish must be other than as specified (see 3.5.).
e. Inspection conditions, if other than as specified (see 4.2).
f. Sample size for first article inspection (see 4.3).
g. Sample size for examination and acceptance tests (see 4.4.1).
h. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 3.2.5 and 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.3.3 Test temperature stabilization. Test temperature stabilization is the point at which the test sample will not vary more than 5°F (3°C) per hour (see 4.6.2.6 and 4.6.2.7).

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6.4 Subject term (key word) listing.

Chromium
Pump assembly
Pump, displacement
Pump, double acting
Pump, hand operated

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2520-0001)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-45307C(AT)	961113

3. DOCUMENT TITLE

Pump, Hydraulic, Axial Pistons; Manual, With Fire-Control Switch

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(810) 574-8745	786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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